

Module Title: Knowledge Representation
and Reasoning

© **UNIVERSITY OF LEEDS**

School of Computing

Semester 1 2019/20

Calculator instructions:

- You **are** allowed to use a non-programmable calculator only from the following list of approved models in this examination: **Casio FX-82, Casio FX-83, Casio FX-85.**

Dictionary instructions:

- A basic English dictionary is available to use: raise your hand and ask an invigilator, if you need it.

Examination Information

- There are **5** pages to this examination.
- There are **2 hours** to complete the examination.
- Answer **all 3** questions.
- The number in brackets [] indicates the marks available for each question or part question.
- You are reminded of the need for clear presentation in your answers.
- The total number of marks for this examination paper is **60**.
- You are allowed to use annotated materials.

Please do not remove this paper from the exam venue.

Question 1

- (a) Translate the following sentence into *Propositional Logic*: [2 marks]

I go to the park on Sundays unless it is raining.

- (b) Translate the following sentences into *First-Order Predicate Logic* (using equality where necessary):

(i) John found a green beetle. [2 marks]

(ii) Rabbits don't like chocolate, except for some old rabbits. [2 marks]

(iii) All friends of Mary know each other. [2 marks]

(iv) One of my uncles does not like any of his own children. [2 marks]

- (c) Give an English sentence that captures the meaning of the following formula of first-order logic in a concise and natural way: [2 marks]

$$\begin{aligned} \forall x [& \text{Bicycle}(x) \rightarrow \\ & \exists y \exists z [\neg(y = z) \wedge \text{Wheel}(y) \wedge \text{Wheel}(z) \wedge \\ & \text{HasPart}(x, y) \wedge \text{HasPart}(x, z) \wedge \\ & \forall w [(\text{HasPart}(x, w) \wedge \text{Wheel}(w)) \rightarrow (w = y \vee w = z)]]] \end{aligned}$$

- (d) $\mathcal{M} = \langle \mathcal{D}, \delta \rangle$ is a model for a first-order language with two unary predicates P and Q and a binary relation predicate R . The domain of $\mathcal{M}$ is the set $\{a, b, c, d, e\}$, and the denotation of the predicates is:

- $\delta(P) = \{a, b, c\}$
- $\delta(Q) = \{c, d, e\}$
- $\delta(R) = \{\langle a, d \rangle, \langle b, e \rangle, \langle c, c \rangle\}$

Which of the following formulae are satisfied by this model? [2 marks]

F1. $\exists x [P(x) \wedge Q(x)]$

F2. $\exists x \exists y [\neg(x = y) \wedge \forall z [\neg R(z, x) \wedge \neg R(z, y)]]$

F3. $\forall x [P(x) \rightarrow \exists y [R(x, y) \wedge Q(y)]]$

F4. $\neg \exists x \exists y [Q(x) \wedge Q(y) \wedge R(x, y)]$

(e) Use the *Sequent Calculus* to show that the following sequent is valid:

[6 marks]

$$\forall x[P(x) \wedge Q(x)], (P(a) \wedge Q(b)) \rightarrow R \vdash R$$

You should only use rules from the following rule set, which was presented in the lecture slides, to construct your proof:

$$\frac{\text{Axiom}}{\alpha, \Gamma \vdash \alpha, \Delta}$$

$$\frac{\alpha, \beta, \Gamma \vdash \Delta}{(\alpha \wedge \beta), \Gamma \vdash \Delta} [\wedge\vdash]$$

$$\frac{\Gamma \vdash \alpha, \Delta \quad \text{and} \quad \Gamma \vdash \beta, \Delta}{\Gamma \vdash (\alpha \wedge \beta), \Delta} [\vdash \wedge]$$

$$\frac{\alpha, \Gamma \vdash \Delta \quad \text{and} \quad \beta, \Gamma \vdash \Delta}{(\alpha \vee \beta), \Gamma \vdash \Delta} [\vee\vdash]$$

$$\frac{\Gamma \vdash \alpha, \beta, \Delta}{\Gamma \vdash (\alpha \vee \beta), \Delta} [\vdash \vee]$$

$$\frac{\Gamma \vdash \alpha, \Delta}{\neg\alpha, \Gamma \vdash \Delta} [\neg\vdash]$$

$$\frac{\Gamma, \alpha \vdash \Delta}{\Gamma \vdash \neg\alpha, \Delta} [\vdash \neg]$$

$$\frac{\Gamma, \neg\alpha \vee \beta \vdash \Delta}{\Gamma, \alpha \rightarrow \beta \vdash \Delta} [\rightarrow \vdash r.w.]$$

$$\frac{\Gamma \vdash \neg\alpha \vee \beta, \Delta}{\Gamma \vdash \alpha \rightarrow \beta, \Delta} [\vdash \rightarrow r.w.]$$

$$\frac{\forall x[\Phi(x)], \Phi(k), \Gamma \vdash \Delta}{\forall x[\Phi(x)], \Gamma \vdash \Delta} [\forall\vdash]$$

$$\frac{\Gamma \vdash \Phi(k), \Delta}{\Gamma \vdash \forall x[\Phi(x)], \Delta} [\vdash \forall]^\dagger$$

† where κ cannot occur anywhere in the lower sequent.

[Question 1 total: 20 marks]

Question 2

- (a) (i) Give the set of *clausal* formulae (i.e. formulae in *conjunctive normal form*) corresponding to the following propositional formulae: **[4 marks]**

$$\neg\neg P, (P \rightarrow Q), ((Q \wedge R) \rightarrow S), \neg(R \rightarrow S)$$

- (ii) Give a proof that these formulae are inconsistent using *binary propositional resolution*. **[4 marks]**

- (b) Translate the following sentence into *Propositional Tense Logic*: **[2 marks]**

I found your wallet after you had left.

- (c) An AI specialist wants to model using *Situation Calculus* a situation in which a group of agents each possess various precious jewels of different colours, which they exchange with each other. Each agent wants to acquire any jewel that is of a colour that they do not possess, and, if they only have one jewel of a particular colour, they want to keep that jewel. Two agents will exchange jewels with each other only if: one of the two receives a jewel that they want to acquire; and neither of them gives a jewel that they want to keep.

The only *fluent* used in the representation will be:

- $\text{has}(a, j)$ — agent a has jewel j .

The following predicate will also be used:

- $\text{Colour}(j, c)$ — jewel j is of colour c .

(This is a normal predicate, not a fluent, since the colour of a jewel cannot change.)

The crucial part of the Situation Calculus representation will be axiomatisation of the action of two agents exchanging jewels. This action will be represented as follows:

- $\text{Exchange}(a_1, j_1, a_2, j_2)$ — agent a_1 gives jewel j_1 to agent a_2 ,
and agent a_2 gives jewel j_2 to agent a_1 .

- (i) Give an *effect* axiom specifying all changes of fluents caused by the **Exchange** action. **[3 marks]**

- (ii) Give an appropriate *pre-condition* axiom for the **Exchange** action, which captures the conditions required for an exchange to take place. **[4 marks]**

- (iii) Give a suitable *frame axiom* which specifies what will stay the same after the **Exchange** action. **[3 marks]**

[Question 2 total: 20 marks]

Question 3

- (a) For each of the following *Prolog* queries, give the value of the variable X after the query has been executed:

(i) $?- Y = 8, Y/2 = X.$ [1 mark]

(ii) $?- [_ | Y] = [\text{which}, \text{one}, \text{is}, \text{it}], [_ , X | _] = Y.$ [1 mark]

(iii) $?- L = [1,2], \text{setof}([A,B], (\text{member}(A,L), \text{member}(B,L))), X).$ [2 marks]

- (b) Consider the following formulae involving topological relations of the *Region Connection Calculus* (RCC) and the *convex hull* function, *conv*. The constants (a, b and c) refer to particular spatial regions. In each case, draw a configuration of the regions referred to by these constants that satisfies the formula, labelling your diagram to indicate which region is which:

(i) $\text{NTPP}(\text{sum}(a, b), c) \wedge \text{PO}(a, b)$ [2 marks]

(ii) $P(a, b) \wedge P(b, c) \wedge \text{TPP}(a, c)$ [2 marks]

(iii) $\text{DC}(a, b) \wedge \text{PO}(a, \text{conv}(b))$ [2 marks]

- (c) Represent the following statements in *Description Logic*:

(i) Humans are a kind of non-aquatic mammal. [2 marks]

(ii) Every happy child has a happy friend. [2 marks]

- (d) Give an explanation in English of the following *Default Logic* rule: [2 marks]

$$\text{Friend}(x, y) \wedge \text{Friend}(x, z) : \text{Friend}(y, z) / \text{Friend}(y, z)$$

- (e) This question concerns a *Fuzzy Logic* in which the following definitions of truth functions for *linguistic modifiers* are specified:

$$\text{quite}(\phi) = \phi^{\frac{1}{2}} \quad \text{very}(\phi) = \phi^2$$

The logic is used to describe Jumbo the elephant, who possesses certain characteristics to the following degrees:

$$\text{Large}(\text{jumbo}) = 0.7 \quad \text{Intelligent}(\text{jumbo}) = 0.36$$

Translate the following sentences into fuzzy logic and also give the fuzzy truth value of each proposition (under the standard fuzzy interpretation of the Boolean connectives):

(i) Jumbo is quite intelligent. [2 marks]

(ii) Jumbo is intelligent and not very large. [2 marks]

[Question 3 total: 20 marks]

[Grand total: 60 marks]